

Global oases diverge in social concentration, atmospheric water demand and thermal exposure across desert climates

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Supplementary information

Supplementary Table 1 | Effect sizes and statistical inference for 31 prespecified comparisons between oases in hot-desert and cold-desert climates

Domain	Metric	Effect scale	Effect	95% interval	FDR q	Sample size, n (BWh/BWk)	Two-sided plus-one p
Social exposure	GHS-POP 2020 oasis population density	Log ratio of group medians	1.15	0.31 to 1.83	0.012	1,822/742	0.0120
Social exposure	GHS-BUILT-S 2020 built-up-area fraction	Difference in medians (proportion)	0.0037	0.0016 to 0.0060	0.002	1,822/742	0.0010
Water conditions	Background aridity index (AI)	Difference in area-weighted means (index)	-0.025	-0.078 to 0.027	0.615	1,822/742	0.3688
Water conditions	TerraClimate 1991–2020 climatic water deficit	Difference in area-weighted means (mm yr ⁻¹)	726.55	586.75 to 912.24	0.002	1,822/742	0.0010
Water conditions	JRC 1984–2021 any observed surface-water fraction	Difference in area-weighted means (proportion)	-0.012	-0.069 to 0.033	0.801	1,822/742	0.6407
Thermal exposure	Annual days with maximum UTCI ≥32 °C (oasis median)	Difference in medians (d yr ⁻¹)	140.03	109.73 to 165.34	<0.001	1,822/742	0.0010
Thermal exposure	Annual days with minimum UTCI ≤-13 °C (oasis median)	Difference in medians (d yr ⁻¹)	-123.77	-153.47 to -71.46	<0.001	1,822/742	0.0010
Thermal exposure	Annual days with maximum UTCI ≥32 °C (2020 population weighted)	Difference in population-weighted means (d yr ⁻¹)	138.03	100.34 to 177.24	<0.001	1,822/742	0.0010
Thermal exposure	Annual days with minimum UTCI ≤-13 °C (2020 population weighted)	Difference in population-weighted means (d yr ⁻¹)	-99.67	-121.90 to -82.84	<0.001	1,822/742	0.0010
Future background climate	NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2041–2070 relative to 1995–2014	Difference in medians (°C)	-0.18	-0.27 to -0.11	0.002	1,822/742	0.0010
Future background climate	NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2041–2070 relative to 1995–2014	Difference in medians (°C)	-0.16	-0.27 to -0.05	0.007	1,822/742	0.0050
Future background climate	NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2041–2070 relative to 1995–2014	Difference in medians (mm d ⁻¹)	-0.030	-0.043 to -0.015	0.002	1,822/742	0.0010
Future background climate	NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2071–2099 relative to 1995–2014	Difference in medians (°C)	-0.24	-0.34 to -0.12	0.002	1,822/742	0.0010
Future background climate	NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2071–2099 relative to 1995–2014	Difference in medians (°C)	-0.23	-0.36 to -0.03	0.013	1,822/742	0.0110
Future background climate	NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2071–2099 relative to 1995–2014	Difference in medians (mm d ⁻¹)	-0.046	-0.060 to -0.028	0.002	1,822/742	0.0010
Future background climate	NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2041–2070 relative to 1995–2014	Difference in medians (°C)	-0.23	-0.35 to -0.12	0.003	1,822/742	0.0020
Future background climate	NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2041–2070 relative to 1995–2014	Difference in medians (°C)	-0.22	-0.35 to -0.02	0.034	1,822/742	0.0310
Future background climate	NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2041–2070 relative to 1995–2014	Difference in medians (mm d ⁻¹)	-0.044	-0.058 to -0.022	0.002	1,822/742	0.0010
Future background climate	NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2071–2099 relative to 1995–2014	Difference in medians (°C)	-0.52	-0.71 to -0.33	0.002	1,822/742	0.0010
Future background climate	NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2071–2099 relative to 1995–2014	Difference in medians (°C)	-0.23	-0.52 to 0.03	0.112	1,822/742	0.1119
Future background climate	NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2071–2099 relative to 1995–2014	Difference in medians (mm d ⁻¹)	-0.08	-0.10 to -0.05	0.002	1,822/742	0.0010
Future population	FuturePop SSP2 oasis population, 2050	Difference in medians (people)	887.62	440.56 to 1452.98	0.003	1,822/742	0.0030
Future population	FuturePop SSP2 oasis population, 2080	Difference in medians (people)	1052.04	575.19 to 1664.89	0.002	1,822/742	0.0010
Future population	FuturePop SSP5 oasis population, 2050	Difference in medians (people)	829.91	385.37 to 1324.81	0.003	1,822/742	0.0020
Future population	FuturePop SSP5 oasis population, 2080	Difference in medians (people)	868.66	409.12 to 1332.60	0.002	1,822/742	0.0010
Economic and night-time light	Kummu GDP v4 2020 total GDP	Log ratio of group medians	0.68	0.02 to 1.30	0.057	1,822/742	0.0430

Domain	Metric	Effect scale	Effect	95% interval	FDR q	Sample size, n (BWh/BWk)	Two-sided plus-one p
Economic and night-time light	Kummu GDP v4 2020 GDP per capita	Log ratio of group medians	−0.35	−0.70 to 0.07	0.139	1,822/742	0.1389
Economic and night-time light	VIIRS Annual VNL v2.1 mean radiance	Log ratio of group medians	1.52	0.63 to 2.34	0.004	1,822/742	0.0010
Economic and night-time light	VIIRS Annual VNL v2.1 lit-area fraction	Log ratio of group medians	0.97	0.37 to 1.59	0.004	1,822/742	0.0020
Water conditions	Global AI/ET0 v3.1 ET0	Difference in area-weighted means (mm yr ^{−1})	806.79	662.68 to 986.12	0.002	1,822/742	0.0010
Water conditions	Global AI/ET0 v3.1 ET0-SD	Difference in area-weighted means (mm yr ^{−1})	−0.80	−15.36 to 13.80	0.849	1,822/742	0.8486

Note: The 95% intervals and two-sided plus-one p values were derived from 2,000 valid replicates of a 500-km spatial-block bootstrap (148 occupied blocks; n(BWh)=1,822, n(BWk)=742). FDR q values were adjusted separately within the six prespecified domains using the Benjamini–Hochberg procedure. Effects are expressed as oases in hot-desert climates minus oases in cold-desert climates.

Supplementary Table 2 | Sensitivity of the primary comparisons to spatial scale, estimand specification and sample treatment

Panel A | Spatial-block scale

Metric	Scale (km)	Blocks	Valid replicates	n (BWh/BWk)	Effect	95% interval	p	q
GHS-POP 2020 oasis population density	500	148	2000	1,822/742	1.15	0.312–1.83	0.0120	0.0120
GHS-BUILT-S 2020 built-up-area fraction	500	148	2000	1,822/742	0.0037	0.0016–0.0060	0.0010	0.0020
Background aridity index (AI)	500	148	2000	1,822/742	-0.0248	-0.0784–0.0266	0.3688	0.6147
TerraClimate 1991–2020 climatic water deficit	500	148	2000	1,822/742	726.6	586.8–912.2	0.0010	0.0025
JRC 1984–2021 any observed surface-water fraction	500	148	2000	1,822/742	-0.0123	-0.0688–0.0331	0.6407	0.8008
Annual days with maximum UTCI ≥ 32 °C (oasis median)	500	148	2000	1,822/742	140.0	109.7–165.3	0.0010	0.0010
Annual days with minimum UTCI ≤ -13 °C (oasis median)	500	148	2000	1,822/742	-123.8	-153.5–-71.46	0.0010	0.0010
Annual days with maximum UTCI ≥ 32 °C (2020 population weighted)	500	148	2000	1,822/742	138.0	100.3–177.2	0.0010	0.0010
Annual days with minimum UTCI ≤ -13 °C (2020 population weighted)	500	148	2000	1,822/742	-99.67	-121.9–-82.84	0.0010	0.0010
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2041–2070 relative to 1995–2014	500	148	2000	1,822/742	-0.179	-0.270–-0.109	0.0010	0.0017
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2041–2070 relative to 1995–2014	500	148	2000	1,822/742	-0.163	-0.270–-0.0485	0.0050	0.0067
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2041–2070 relative to 1995–2014	500	148	2000	1,822/742	-0.0299	-0.0426–-0.0151	0.0010	0.0017
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2071–2099 relative to 1995–2014	500	148	2000	1,822/742	-0.237	-0.339–-0.116	0.0010	0.0017
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2071–2099 relative to 1995–2014	500	148	2000	1,822/742	-0.227	-0.365–-0.0319	0.0110	0.0132
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2071–2099 relative to 1995–2014	500	148	2000	1,822/742	-0.0460	-0.0599–-0.0277	0.0010	0.0017
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2041–2070 relative to 1995–2014	500	148	2000	1,822/742	-0.232	-0.354–-0.119	0.0020	0.0030
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2041–2070 relative to 1995–2014	500	148	2000	1,822/742	-0.219	-0.347–-0.0187	0.0310	0.0338
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2041–2070 relative to 1995–2014	500	148	2000	1,822/742	-0.0441	-0.0577–-0.0223	0.0010	0.0017
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2071–2099 relative to 1995–2014	500	148	2000	1,822/742	-0.525	-0.706–-0.331	0.0010	0.0017
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2071–2099 relative to 1995–2014	500	148	2000	1,822/742	-0.234	-0.523–-0.0313	0.1119	0.1119
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2071–2099 relative to 1995–2014	500	148	2000	1,822/742	-0.0816	-0.103–-0.0493	0.0010	0.0017
FuturePop SSP2 oasis population, 2050	500	148	2000	1,822/742	887.6	440.6–1453.0	0.0030	0.0030
FuturePop SSP2 oasis population, 2080	500	148	2000	1,822/742	1052.0	575.2–1664.9	0.0010	0.0020
FuturePop SSP5 oasis population, 2050	500	148	2000	1,822/742	829.9	385.4–1324.8	0.0020	0.0027
FuturePop SSP5 oasis population, 2080	500	148	2000	1,822/742	868.7	409.1–1332.6	0.0010	0.0020
Kummu GDP v4 2020 total GDP	500	148	2000	1,822/742	0.680	0.0238–1.30	0.0430	0.0573
Kummu GDP v4 2020 GDP per capita	500	148	2000	1,822/742	-0.347	-0.695–-0.0695	0.1389	0.1389
VIIRS Annual VNL v2.1 mean radiance	500	148	2000	1,822/742	1.52	0.635–2.34	0.0010	0.0040
VIIRS Annual VNL v2.1 lit-area fraction	500	148	2000	1,822/742	0.972	0.370–1.59	0.0020	0.0040
Global AI/ET0 v3.1 ET0	500	148	2000	1,822/742	806.8	662.7–986.1	0.0010	0.0025
Global AI/ET0 v3.1 ET0-SD	500	148	2000	1,822/742	-0.796	-15.36–-13.80	0.8486	0.8486
GHS-POP 2020 oasis population density	250	352	1000	1,822/742	1.15	0.428–1.74	0.0040	0.0040
GHS-POP 2020 oasis population density	1000	65	1000	1,822/742	1.15	0.179–1.88	0.0240	0.0240
GHS-BUILT-S 2020 built-up-area fraction	250	352	1000	1,822/742	0.0037	0.0021–0.0057	0.0020	0.0040
GHS-BUILT-S 2020 built-up-area fraction	1000	65	1000	1,822/742	0.0037	0.0008–0.0069	0.0180	0.0240
Background aridity index (AI)	250	352	1000	1,822/742	-0.0248	-0.0729–0.0252	0.3317	0.5528
Background aridity index (AI)	1000	65	1000	1,822/742	-0.0248	-0.0851–-0.0269	0.3137	0.5228
TerraClimate 1991–2020 climatic water deficit	250	352	1000	1,822/742	726.6	596.7–905.4	0.0020	0.0050
TerraClimate 1991–2020 climatic water deficit	1000	65	1000	1,822/742	726.6	574.4–931.4	0.0020	0.0050
JRC 1984–2021 any observed surface-water fraction	250	352	1000	1,822/742	-0.0123	-0.0621–-0.0319	0.6454	0.8067
JRC 1984–2021 any observed surface-water fraction	1000	65	1000	1,822/742	-0.0123	-0.0742–-0.0355	0.6414	0.8017
Annual days with maximum UTCI ≥ 32 °C (oasis median)	250	352	1000	1,822/742	140.0	117.1–157.3	0.0020	0.0020
Annual days with maximum UTCI ≥ 32 °C (oasis median)	1000	65	1000	1,822/742	140.0	102.3–168.2	0.0020	0.0020

Metric	Scale (km)	Blocks	Valid replicates	n (BWh/BWk)	Effect	95% interval	p	q
Annual days with minimum UTCI ≤−13 °C (oasis median)	250	352	1000	1,822/742	-123.8	-143.8—88.22	0.0020	0.0020
Annual days with minimum UTCI ≤−13 °C (oasis median)	1000	65	1000	1,822/742	-123.8	-154.2—54.06	0.0020	0.0020
Annual days with maximum UTCI ≥32 °C (2020 population weighted)	250	352	1000	1,822/742	138.0	101.3—173.6	0.0020	0.0020
Annual days with maximum UTCI ≥32 °C (2020 population weighted)	1000	65	1000	1,822/742	138.0	99.30—183.4	0.0020	0.0020
Annual days with minimum UTCI ≤−13 °C (2020 population weighted)	250	352	1000	1,822/742	-99.67	-120.5—85.01	0.0020	0.0020
Annual days with minimum UTCI ≤−13 °C (2020 population weighted)	1000	65	1000	1,822/742	-99.67	-123.5—80.87	0.0020	0.0020
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2041–2070 relative to 1995–2014	250	352	1000	1,822/742	-0.179	-0.241—0.128	0.0020	0.0024
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2041–2070 relative to 1995–2014	1000	65	1000	1,822/742	-0.179	-0.306—0.0934	0.0020	0.0040
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2041–2070 relative to 1995–2014	250	352	1000	1,822/742	-0.163	-0.244—0.0911	0.0020	0.0024
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2041–2070 relative to 1995–2014	1000	65	1000	1,822/742	-0.163	-0.307—0.0008	0.0480	0.0639
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2041–2070 relative to 1995–2014	250	352	1000	1,822/742	-0.0299	-0.0394—0.0179	0.0020	0.0024
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2041–2070 relative to 1995–2014	1000	65	1000	1,822/742	-0.0299	-0.0439—0.0110	0.0040	0.0060
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2071–2099 relative to 1995–2014	250	352	1000	1,822/742	-0.237	-0.308—0.152	0.0020	0.0024
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2071–2099 relative to 1995–2014	1000	65	1000	1,822/742	-0.237	-0.403—0.0908	0.0020	0.0040
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2071–2099 relative to 1995–2014	250	352	1000	1,822/742	-0.227	-0.328—0.0863	0.0020	0.0024
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2071–2099 relative to 1995–2014	1000	65	2000	1,822/742	-0.227	-0.427—0.0140	0.0740	0.0829
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2071–2099 relative to 1995–2014	250	352	1000	1,822/742	-0.0460	-0.0591—0.0338	0.0020	0.0024
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2071–2099 relative to 1995–2014	1000	65	1000	1,822/742	-0.0460	-0.0633—0.0217	0.0040	0.0060
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2041–2070 relative to 1995–2014	250	352	1000	1,822/742	-0.232	-0.325—0.153	0.0020	0.0024
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2041–2070 relative to 1995–2014	1000	65	1000	1,822/742	-0.232	-0.397—0.106	0.0020	0.0040
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2041–2070 relative to 1995–2014	250	352	1000	1,822/742	-0.219	-0.310—0.0832	0.0060	0.0065
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2041–2070 relative to 1995–2014	1000	65	2000	1,822/742	-0.219	-0.420—0.0136	0.0760	0.0829
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2041–2070 relative to 1995–2014	250	352	1000	1,822/742	-0.0441	-0.0564—0.0287	0.0020	0.0024
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2041–2070 relative to 1995–2014	1000	65	1000	1,822/742	-0.0441	-0.0606—0.0186	0.0020	0.0040
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2071–2099 relative to 1995–2014	250	352	1000	1,822/742	-0.525	-0.637—0.382	0.0020	0.0024
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2071–2099 relative to 1995–2014	1000	65	1000	1,822/742	-0.525	-0.807—0.300	0.0020	0.0040
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2071–2099 relative to 1995–2014	250	352	2000	1,822/742	-0.234	-0.425—0.0121	0.0240	0.0240
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2071–2099 relative to 1995–2014	1000	65	1000	1,822/742	-0.234	-0.773—0.0668	0.2697	0.2697
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2071–2099 relative to 1995–2014	250	352	1000	1,822/742	-0.0816	-0.0992—0.0579	0.0020	0.0024
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2071–2099 relative to 1995–2014	1000	65	1000	1,822/742	-0.0816	-0.109—0.0465	0.0020	0.0040
FuturePop SSP2 oasis population, 2050	250	352	1000	1,822/742	887.6	579.5—1246.8	0.0020	0.0020
FuturePop SSP2 oasis population, 2050	1000	65	1000	1,822/742	887.6	358.1—1520.6	0.0040	0.0053
FuturePop SSP2 oasis population, 2080	250	352	1000	1,822/742	1052.0	753.4—1486.8	0.0020	0.0020
FuturePop SSP2 oasis population, 2080	1000	65	1000	1,822/742	1052.0	468.1—1883.5	0.0040	0.0053
FuturePop SSP5 oasis population, 2050	250	352	1000	1,822/742	829.9	522.4—1139.9	0.0020	0.0020
FuturePop SSP5 oasis population, 2050	1000	65	1000	1,822/742	829.9	296.4—1428.0	0.0060	0.0060
FuturePop SSP5 oasis population, 2080	250	352	1000	1,822/742	868.7	570.7—1233.7	0.0020	0.0020
FuturePop SSP5 oasis population, 2080	1000	65	1000	1,822/742	868.7	288.1—1561.3	0.0040	0.0053
Kummu GDP v4 2020 total GDP	250	352	1000	1,822/742	0.680	0.0786—1.23	0.0240	0.0320

Metric	Scale (km)	Blocks	Valid replicates	n (BWh/BWk)	Effect	95% interval	p	q
Kummu GDP v4 2020 total GDP	1000	65	2000	1,822/742	0.680	-0.123–1.39	0.0860	0.1146
Kummu GDP v4 2020 GDP per capita	250	352	2000	1,822/742	-0.347	-0.594–0.0013	0.0540	0.0540
Kummu GDP v4 2020 GDP per capita	1000	65	1000	1,822/742	-0.347	-0.685–0.171	0.1978	0.1978
VIIRS Annual VNL v2.1 mean radiance	250	352	1000	1,822/742	1.52	0.888–2.17	0.0020	0.0040
VIIRS Annual VNL v2.1 mean radiance	1000	65	1000	1,822/742	1.52	0.269–2.30	0.0260	0.0679
VIIRS Annual VNL v2.1 lit-area fraction	250	352	1000	1,822/742	0.972	0.543–1.50	0.0020	0.0040
VIIRS Annual VNL v2.1 lit-area fraction	1000	65	1000	1,822/742	0.972	0.0712–1.57	0.0340	0.0679
Global AI/ET0 v3.1 ET0	250	352	1000	1,822/742	806.8	663.3–983.8	0.0020	0.0050
Global AI/ET0 v3.1 ET0	1000	65	1000	1,822/742	806.8	654.7–1000.3	0.0020	0.0050
Global AI/ET0 v3.1 ET0-SD	250	352	1000	1,822/742	-0.796	-14.27–14.48	0.8372	0.8372
Global AI/ET0 v3.1 ET0-SD	1000	65	1000	1,822/742	-0.796	-17.67–14.14	0.9171	0.9171

Panel B | Prespecified method and sample treatments

Metric	Analysis	n (BWh/BWk)	Valid replicates	Effect	95% interval	p	q
GHS-POP 2020 oasis population density	Equal weights	1,822/742	2000	1.15	0.280–1.83	0.0070	0.0070
GHS-POP 2020 oasis population density	Area weights	1,822/742	2000	0.188	-0.350–1.85	0.2969	0.5937
GHS-POP 2020 oasis population density	Population weights	1,822/742	2000	1.08	0.0409–1.88	0.0370	0.0740
GHS-POP 2020 oasis population density	C_swap	1,822/742	2000	0.658	0.283–1.09	0.0030	0.0030
GHS-POP 2020 oasis population density	Interpolated median	1,822/742	2000	1.15	0.268–1.78	0.0090	0.0090
GHS-POP 2020 oasis population density	Upper median	1,822/742	2000	1.15	0.314–1.81	0.0040	0.0040
GHS-BUILT-S 2020 built-up-area fraction	Equal weights	1,822/742	2000	0.0037	0.0016–0.0059	0.0020	0.0040
GHS-BUILT-S 2020 built-up-area fraction	Area weights	1,822/742	2000	-0.0052	-0.0169–0.0106	0.8506	0.8506
GHS-BUILT-S 2020 built-up-area fraction	Population weights	1,822/742	2000	-0.0058	-0.0147–0.0262	0.9825	0.9825
GHS-BUILT-S 2020 built-up-area fraction	C_swap	1,822/742	2000	0.0089	0.0043–0.0142	0.0010	0.0020
GHS-BUILT-S 2020 built-up-area fraction	Interpolated median	1,822/742	2000	0.0037	0.0016–0.0060	0.0030	0.0060
GHS-BUILT-S 2020 built-up-area fraction	Upper median	1,822/742	2000	0.0037	0.0017–0.0061	0.0020	0.0040
Background aridity index (AI)	Equal weights	1,822/742	2000	-0.0391	-0.0589–0.0220	0.0010	0.0017
Background aridity index (AI)	Area weights	1,822/742	2000	-0.0248	-0.0763–0.0260	0.3428	0.5714
Background aridity index (AI)	Population weights	1,822/742	2000	-0.0558	-0.118–0.0160	0.1179	0.1966
Background aridity index (AI)	C_swap	1,822/742	2000	0.0027	-0.120–0.0758	0.7956	0.7956
TerraClimate 1991–2020 climatic water deficit	Equal weights	1,822/742	2000	824.3	685.0–957.1	0.0010	0.0017
TerraClimate 1991–2020 climatic water deficit	Area weights	1,822/742	2000	726.6	580.8–913.9	0.0010	0.0025
TerraClimate 1991–2020 climatic water deficit	Population weights	1,822/742	2000	629.0	465.4–884.1	0.0010	0.0025
TerraClimate 1991–2020 climatic water deficit	C_swap	1,822/742	2000	723.8	390.2–931.0	0.0010	0.0025
JRC 1984–2021 any observed surface-water fraction	Equal weights	1,822/742	2000	-0.0126	-0.0251–0.0011	0.0380	0.0475
JRC 1984–2021 any observed surface-water fraction	Area weights	1,822/742	2000	-0.0123	-0.0674–0.0343	0.6677	0.8346
JRC 1984–2021 any observed surface-water fraction	Population weights	1,822/742	2000	-0.0136	-0.0723–0.0402	0.5357	0.5357
JRC 1984–2021 any observed surface-water fraction	C_swap	1,822/742	2000	-0.0079	-0.0826–0.0623	0.7726	0.7956
Annual days with maximum UTCI ≥32 °C (oasis median)	Equal weights	1,822/742	2000	140.0	107.6–165.4	0.0010	0.0010
Annual days with maximum UTCI ≥32 °C (oasis median)	Area weights	1,822/742	2000	114.2	91.53–196.5	0.0010	0.0010
Annual days with maximum UTCI ≥32 °C (oasis median)	Population weights	1,822/742	2000	135.1	75.50–197.8	0.0010	0.0010
Annual days with maximum UTCI ≥32 °C (oasis median)	C_swap	1,822/742	2000	139.3	118.2–159.1	0.0010	0.0010
Annual days with maximum UTCI ≥32 °C (oasis median)	Interpolated median	1,822/742	2000	139.8	109.4–165.3	0.0010	0.0010
Annual days with maximum UTCI ≥32 °C (oasis median)	Upper median	1,822/742	2000	139.5	109.3–166.5	0.0010	0.0010
Annual days with minimum UTCI ≤–13 °C (oasis median)	Equal weights	1,822/742	2000	-123.8	-152.8–70.20	0.0010	0.0010
Annual days with minimum UTCI ≤–13 °C (oasis median)	Area weights	1,822/742	2000	-114.4	-140.2–79.63	0.0010	0.0010
Annual days with minimum UTCI ≤–13 °C (oasis median)	Population weights	1,822/742	2000	-88.93	-140.1–78.70	0.0010	0.0010
Annual days with minimum UTCI ≤–13 °C (oasis median)	C_swap	1,822/742	2000	-111.6	-133.1–84.55	0.0010	0.0010
Annual days with minimum UTCI ≤–13 °C (oasis median)	Interpolated median	1,822/742	2000	-124.5	-154.6–70.30	0.0010	0.0010
Annual days with minimum UTCI ≤–13 °C (oasis median)	Upper median	1,822/742	2000	-125.3	-152.2–71.39	0.0010	0.0010
Annual days with maximum UTCI ≥32 °C (2020 population weighted)	Equal weights	1,822/742	2000	139.3	119.1–159.1	0.0010	0.0010
Annual days with maximum UTCI ≥32 °C (2020 population weighted)	Area weights	1,822/742	2000	150.6	122.0–184.2	0.0010	0.0010
Annual days with maximum UTCI ≥32 °C (2020 population weighted)	Population weights	1,822/742	2000	138.0	98.28–175.4	0.0010	0.0010

Metric	Analysis	n (BWh/BWk)	Valid replicates	Effect	95% interval	p	q
Annual days with maximum UTCI ≥ 32 °C (2020 population weighted)	C_swap	1,822/742	2000	135.1	75.50–198.4	0.0010	0.0010
Annual days with minimum UTCI ≤ -13 °C (2020 population weighted)	Equal weights	1,822/742	2000	-111.6	-132.9–84.54	0.0010	0.0010
Annual days with minimum UTCI ≤ -13 °C (2020 population weighted)	Area weights	1,822/742	2000	-112.1	-138.4–93.85	0.0010	0.0010
Annual days with minimum UTCI ≤ -13 °C (2020 population weighted)	Population weights	1,822/742	2000	-99.67	-121.0–82.92	0.0010	0.0010
Annual days with minimum UTCI ≤ -13 °C (2020 population weighted)	C_swap	1,822/742	2000	-88.93	-140.1–78.20	0.0010	0.0010
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2041–2070 relative to 1995–2014	Equal weights	1,822/742	2000	-0.179	-0.270–0.113	0.0010	0.0017
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2041–2070 relative to 1995–2014	Area weights	1,822/742	2000	-0.386	-0.426–0.106	0.1599	0.2799
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2041–2070 relative to 1995–2014	Population weights	1,822/742	2000	-0.394	-0.426–0.0309	0.0610	0.1049
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2041–2070 relative to 1995–2014	C_swap	1,822/742	2000	-0.216	-0.286–0.151	0.0010	0.0020
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2041–2070 relative to 1995–2014	Interpolated median	1,822/742	2000	-0.179	-0.269–0.106	0.0010	0.0015
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2041–2070 relative to 1995–2014	Upper median	1,822/742	2000	-0.179	-0.267–0.114	0.0010	0.0015
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2041–2070 relative to 1995–2014	Equal weights	1,822/742	2000	-0.163	-0.267–0.0377	0.0060	0.0080
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2041–2070 relative to 1995–2014	Area weights	1,822/742	2000	-0.263	-0.355–0.0353	0.0010	0.0120
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2041–2070 relative to 1995–2014	Population weights	1,822/742	2000	-0.268	-0.422–0.0919	0.0020	0.0240
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2041–2070 relative to 1995–2014	C_swap	1,822/742	2000	-0.183	-0.292–0.0741	0.0010	0.0020
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2041–2070 relative to 1995–2014	Interpolated median	1,822/742	2000	-0.163	-0.265–0.0451	0.0150	0.0180
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2041–2070 relative to 1995–2014	Upper median	1,822/742	2000	-0.163	-0.267–0.0410	0.0080	0.0107
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2041–2070 relative to 1995–2014	Equal weights	1,822/742	2000	-0.0299	-0.0423–0.0139	0.0010	0.0017
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2041–2070 relative to 1995–2014	Area weights	1,822/742	2000	-0.0381	-0.0750–0.0716	0.2699	0.2799
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2041–2070 relative to 1995–2014	Population weights	1,822/742	2000	-0.0387	-0.0908–0.0732	0.6787	0.7256
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2041–2070 relative to 1995–2014	C_swap	1,822/742	2000	-0.0209	-0.0327–0.0080	0.0030	0.0036
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2041–2070 relative to 1995–2014	Interpolated median	1,822/742	2000	-0.0300	-0.0425–0.0148	0.0010	0.0015
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2041–2070 relative to 1995–2014	Upper median	1,822/742	2000	-0.0300	-0.0425–0.0150	0.0010	0.0015
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2071–2099 relative to 1995–2014	Equal weights	1,822/742	2000	-0.237	-0.347–0.115	0.0010	0.0017
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2071–2099 relative to 1995–2014	Area weights	1,822/742	2000	-0.399	-0.530–0.204	0.2269	0.2799
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2071–2099 relative to 1995–2014	Population weights	1,822/742	2000	-0.415	-0.605–0.0295	0.0700	0.1049
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2071–2099 relative to 1995–2014	C_swap	1,822/742	2000	-0.285	-0.379–0.201	0.0010	0.0020
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2071–2099 relative to 1995–2014	Interpolated median	1,822/742	2000	-0.237	-0.339–0.114	0.0010	0.0015
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2071–2099 relative to 1995–2014	Upper median	1,822/742	2000	-0.237	-0.346–0.111	0.0010	0.0015
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2071–2099 relative to 1995–2014	Equal weights	1,822/742	2000	-0.227	-0.362–0.0342	0.0100	0.0120
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2071–2099 relative to 1995–2014	Area weights	1,822/742	2000	-0.243	-0.461–0.0029	0.0480	0.2799
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2071–2099 relative to 1995–2014	Population weights	1,822/742	2000	-0.163	-0.570–0.0508	0.0300	0.1049
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2071–2099 relative to 1995–2014	C_swap	1,822/742	2000	-0.232	-0.366–0.0820	0.0030	0.0036
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2071–2099 relative to 1995–2014	Interpolated median	1,822/742	2000	-0.227	-0.360–0.0345	0.0110	0.0147
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2071–2099 relative to 1995–2014	Upper median	1,822/742	2000	-0.227	-0.368–0.0315	0.0110	0.0132
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2071–2099 relative to 1995–2014	Equal weights	1,822/742	2000	-0.0460	-0.0619–0.0277	0.0010	0.0017
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2071–2099 relative to 1995–2014	Area weights	1,822/742	2000	-0.0535	-0.115–0.101	0.2009	0.2799

Metric	Analysis	n (BWh/BWk)	Valid replicates	Effect	95% interval	p	q
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2071–2099 relative to 1995–2014	Population weights	1,822/742	2000	-0.0590	-0.115–0.110	0.6887	0.7256
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2071–2099 relative to 1995–2014	C_swap	1,822/742	2000	-0.0334	-0.0493–0.0141	0.0050	0.0055
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2071–2099 relative to 1995–2014	Interpolated median	1,822/742	2000	-0.0460	-0.0607–0.0281	0.0010	0.0015
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2071–2099 relative to 1995–2014	Upper median	1,822/742	2000	-0.0460	-0.0616–0.0286	0.0010	0.0015
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2041–2070 relative to 1995–2014	Equal weights	1,822/742	2000	-0.232	-0.357–0.116	0.0020	0.0030
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2041–2070 relative to 1995–2014	Area weights	1,822/742	2000	-0.483	-0.591–0.185	0.1809	0.2799
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2041–2070 relative to 1995–2014	Population weights	1,822/742	2000	-0.520	-0.564–0.0064	0.0460	0.1049
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2041–2070 relative to 1995–2014	C_swap	1,822/742	2000	-0.289	-0.396–0.199	0.0010	0.0020
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2041–2070 relative to 1995–2014	Interpolated median	1,822/742	2000	-0.232	-0.356–0.115	0.0010	0.0015
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2041–2070 relative to 1995–2014	Upper median	1,822/742	2000	-0.231	-0.363–0.118	0.0010	0.0015
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2041–2070 relative to 1995–2014	Equal weights	1,822/742	2000	-0.219	-0.345–0.0286	0.0180	0.0196
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2041–2070 relative to 1995–2014	Area weights	1,822/742	2000	-0.330	-0.522–0.0436	0.0740	0.2799
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2041–2070 relative to 1995–2014	Population weights	1,822/742	2000	-0.291	-0.549–0.0497	0.0380	0.1049
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2041–2070 relative to 1995–2014	C_swap	1,822/742	2000	-0.256	-0.403–0.0924	0.0030	0.0036
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2041–2070 relative to 1995–2014	Interpolated median	1,822/742	2000	-0.219	-0.358–0.0305	0.0190	0.0207
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2041–2070 relative to 1995–2014	Upper median	1,822/742	2000	-0.218	-0.362–0.0321	0.0210	0.0229
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2041–2070 relative to 1995–2014	Equal weights	1,822/742	2000	-0.0441	-0.0585–0.0226	0.0010	0.0017
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2041–2070 relative to 1995–2014	Area weights	1,822/742	2000	-0.0434	-0.0992–0.125	0.2799	0.2799
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2041–2070 relative to 1995–2014	Population weights	1,822/742	2000	-0.0602	-0.106–0.125	0.7256	0.7256
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2041–2070 relative to 1995–2014	C_swap	1,822/742	2000	-0.0261	-0.0443–0.0059	0.0100	0.0100
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2041–2070 relative to 1995–2014	Interpolated median	1,822/742	2000	-0.0441	-0.0579–0.0229	0.0010	0.0015
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2041–2070 relative to 1995–2014	Upper median	1,822/742	2000	-0.0441	-0.0582–0.0227	0.0010	0.0015
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2071–2099 relative to 1995–2014	Equal weights	1,822/742	2000	-0.525	-0.713–0.336	0.0010	0.0017
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2071–2099 relative to 1995–2014	Area weights	1,822/742	2000	-0.984	-1.10–0.116	0.1249	0.2799
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2071–2099 relative to 1995–2014	Population weights	1,822/742	2000	-1.05	-1.13–0.108	0.0390	0.1049
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2071–2099 relative to 1995–2014	C_swap	1,822/742	2000	-0.601	-0.785–0.437	0.0010	0.0020
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2071–2099 relative to 1995–2014	Interpolated median	1,822/742	2000	-0.524	-0.704–0.318	0.0010	0.0015
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2071–2099 relative to 1995–2014	Upper median	1,822/742	2000	-0.524	-0.697–0.338	0.0010	0.0015
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2071–2099 relative to 1995–2014	Equal weights	1,822/742	2000	-0.234	-0.548–0.0315	0.1259	0.1259
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2071–2099 relative to 1995–2014	Area weights	1,822/742	2000	-0.425	-0.866–0.230	0.1599	0.2799
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2071–2099 relative to 1995–2014	Population weights	1,822/742	2000	-0.260	-0.946–0.0386	0.0690	0.1049
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2071–2099 relative to 1995–2014	C_swap	1,822/742	2000	-0.421	-0.710–0.140	0.0010	0.0020
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2071–2099 relative to 1995–2014	Interpolated median	1,822/742	2000	-0.236	-0.602–0.0284	0.1289	0.1289
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2071–2099 relative to 1995–2014	Upper median	1,822/742	2000	-0.237	-0.560–0.0360	0.1449	0.1449
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2071–2099 relative to 1995–2014	Equal weights	1,822/742	2000	-0.0816	-0.103–0.0501	0.0010	0.0017
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2071–2099 relative to 1995–2014	Area weights	1,822/742	2000	-0.104	-0.192–0.186	0.2439	0.2799
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2071–2099 relative to 1995–2014	Population weights	1,822/742	2000	-0.0989	-0.180–0.200	0.6987	0.7256

Metric	Analysis	n (BWh/BWk)	Valid replicates	Effect	95% interval	p	q
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2071–2099 relative to 1995–2014	C_swap	1,822/742	2000	-0.0609	-0.0901–0.0255	0.0030	0.0036
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2071–2099 relative to 1995–2014	Interpolated median	1,822/742	2000	-0.0819	-0.104–0.0502	0.0010	0.0015
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2071–2099 relative to 1995–2014	Upper median	1,822/742	2000	-0.0821	-0.103–0.0506	0.0010	0.0015
FuturePop SSP2 oasis population, 2050	Equal weights	1,822/742	2000	887.6	461.9–1423.6	0.0010	0.0020
FuturePop SSP2 oasis population, 2050	Area weights	1,822/742	2000	9624655.5	-25450096.1–159064379.8	0.3238	0.3728
FuturePop SSP2 oasis population, 2050	Population weights	1,822/742	2000	151735940.6	-25450096.1–261744703.0	0.1449	0.1639
FuturePop SSP2 oasis population, 2050	C_swap	1,822/742	2000	279586.3	-36014.7–715446.0	0.1079	0.1079
FuturePop SSP2 oasis population, 2050	Interpolated median	1,822/742	2000	886.9	429.7–1457.1	0.0010	0.0010
FuturePop SSP2 oasis population, 2050	Upper median	1,822/742	2000	886.1	471.6–1434.7	0.0010	0.0010
FuturePop SSP2 oasis population, 2080	Equal weights	1,822/742	2000	1052.0	584.0–1671.2	0.0010	0.0020
FuturePop SSP2 oasis population, 2080	Area weights	1,822/742	2000	12578744.2	-32396671.3–190061413.3	0.2249	0.3728
FuturePop SSP2 oasis population, 2080	Population weights	1,822/742	2000	183951936.3	-32117911.4–333908811.5	0.0940	0.1639
FuturePop SSP2 oasis population, 2080	C_swap	1,822/742	2000	364942.2	-18589.4–928988.8	0.0740	0.1079
FuturePop SSP2 oasis population, 2080	Interpolated median	1,822/742	2000	1055.0	566.4–1641.2	0.0010	0.0010
FuturePop SSP2 oasis population, 2080	Upper median	1,822/742	2000	1057.9	587.0–1637.6	0.0010	0.0010
FuturePop SSP5 oasis population, 2050	Equal weights	1,822/742	2000	829.9	385.6–1364.7	0.0020	0.0020
FuturePop SSP5 oasis population, 2050	Area weights	1,822/742	2000	8822462.9	-29747997.0–148404593.8	0.3728	0.3728
FuturePop SSP5 oasis population, 2050	Population weights	1,822/742	2000	142350947.3	-26170095.4–230345213.1	0.1639	0.1639
FuturePop SSP5 oasis population, 2050	C_swap	1,822/742	2000	252244.4	-35859.0–638971.1	0.1049	0.1079
FuturePop SSP5 oasis population, 2050	Interpolated median	1,822/742	2000	830.2	371.0–1374.7	0.0010	0.0010
FuturePop SSP5 oasis population, 2050	Upper median	1,822/742	2000	830.5	371.2–1328.7	0.0010	0.0010
FuturePop SSP5 oasis population, 2080	Equal weights	1,822/742	2000	868.7	390.7–1364.5	0.0020	0.0020
FuturePop SSP5 oasis population, 2080	Area weights	1,822/742	2000	10589984.9	-21883795.9–160747634.9	0.2229	0.3728
FuturePop SSP5 oasis population, 2080	Population weights	1,822/742	2000	155027575.4	-24868070.6–247905000.2	0.1029	0.1639
FuturePop SSP5 oasis population, 2080	C_swap	1,822/742	2000	289348.7	-14114.8–698033.5	0.0690	0.1079
FuturePop SSP5 oasis population, 2080	Interpolated median	1,822/742	2000	869.6	413.5–1402.9	0.0010	0.0010
FuturePop SSP5 oasis population, 2080	Upper median	1,822/742	2000	870.6	423.1–1355.7	0.0010	0.0010
Kummu GDP v4 2020 total GDP	Equal weights	1,822/742	2000	0.680	0.0393–1.30	0.0470	0.0626
Kummu GDP v4 2020 total GDP	Area weights	1,822/742	2000	0.754	-2.72–4.17	0.6767	0.6767
Kummu GDP v4 2020 total GDP	Population weights	1,822/742	2000	1.78	-1.29–3.55	0.2299	0.4598
Kummu GDP v4 2020 total GDP	C_swap	1,822/742	2000	0.844	-0.171–1.85	0.1009	0.1346
Kummu GDP v4 2020 total GDP	Interpolated median	1,822/742	2000	0.680	0.0296–1.31	0.0390	0.0520
Kummu GDP v4 2020 total GDP	Upper median	1,822/742	2000	0.680	-0.0000–1.33	0.0510	0.0680
Kummu GDP v4 2020 GDP per capita	Equal weights	1,822/742	2000	-0.347	-0.692–0.0611	0.1339	0.1339
Kummu GDP v4 2020 GDP per capita	Area weights	1,822/742	2000	-0.134	-1.41–0.398	0.5227	0.6767
Kummu GDP v4 2020 GDP per capita	Population weights	1,822/742	2000	0.125	-1.43–0.500	0.7566	0.7566
Kummu GDP v4 2020 GDP per capita	C_swap	1,822/742	2000	-0.187	-0.471–0.115	0.2379	0.2379
Kummu GDP v4 2020 GDP per capita	Interpolated median	1,822/742	2000	-0.347	-0.687–0.0668	0.1499	0.1499
Kummu GDP v4 2020 GDP per capita	Upper median	1,822/742	2000	-0.347	-0.709–0.0666	0.1109	0.1109
VIIRS Annual VNL v2.1 mean radiance	Equal weights	1,822/742	2000	1.52	0.683–2.28	0.0010	0.0040
VIIRS Annual VNL v2.1 mean radiance	Area weights	1,822/742	2000	0.275	-0.303–2.24	0.5097	0.6767
VIIRS Annual VNL v2.1 mean radiance	Population weights	1,822/742	2000	1.16	-0.670–2.23	0.6697	0.7566
VIIRS Annual VNL v2.1 mean radiance	C_swap	1,822/742	2000	1.06	0.545–1.58	0.0010	0.0040
VIIRS Annual VNL v2.1 mean radiance	Interpolated median	1,822/742	2000	1.53	0.683–2.23	0.0030	0.0060
VIIRS Annual VNL v2.1 mean radiance	Upper median	1,822/742	2000	1.53	0.701–2.25	0.0020	0.0080
VIIRS Annual VNL v2.1 lit-area fraction	Equal weights	1,822/742	2000	0.972	0.381–1.61	0.0020	0.0040
VIIRS Annual VNL v2.1 lit-area fraction	Area weights	1,822/742	2000	0.279	-0.149–1.05	0.0960	0.3838
VIIRS Annual VNL v2.1 lit-area fraction	Population weights	1,822/742	2000	0.585	0.132–1.00	0.0110	0.0440
VIIRS Annual VNL v2.1 lit-area fraction	C_swap	1,822/742	2000	0.368	0.149–0.575	0.0030	0.0060
VIIRS Annual VNL v2.1 lit-area fraction	Interpolated median	1,822/742	2000	0.951	0.392–1.60	0.0020	0.0060
VIIRS Annual VNL v2.1 lit-area fraction	Upper median	1,822/742	2000	0.929	0.337–1.57	0.0050	0.0100
Global AI/ET0 v3.1 ET0	Equal weights	1,822/742	2000	852.4	709.4–1004.6	0.0010	0.0017
Global AI/ET0 v3.1 ET0	Area weights	1,822/742	2000	806.8	654.5–984.5	0.0010	0.0025
Global AI/ET0 v3.1 ET0	Population weights	1,822/742	2000	647.9	525.2–939.0	0.0010	0.0025
Global AI/ET0 v3.1 ET0	C_swap	1,822/742	2000	830.8	443.1–1101.3	0.0010	0.0025
Global AI/ET0 v3.1 ET0-SD	Equal weights	1,822/742	2000	-0.918	-8.53–5.80	0.8086	0.8086
Global AI/ET0 v3.1 ET0-SD	Area weights	1,822/742	2000	-0.796	-15.04–15.01	0.8566	0.8566
Global AI/ET0 v3.1 ET0-SD	Population weights	1,822/742	2000	-7.96	-18.56–7.43	0.2939	0.3673
Global AI/ET0 v3.1 ET0-SD	C_swap	1,822/742	2000	-7.98	-20.44–31.90	0.6537	0.7956
GHS-POP 2020 oasis population density	Remove the pooled top 1% by area (26 units)	1,806/732	2000	1.18	0.324–1.89	0.0110	0.0110
GHS-BUILT-S 2020 built-up-area fraction	Remove the pooled top 1% by area (26 units)	1,806/732	2000	0.0036	0.0016–0.0061	0.0040	0.0080
Background aridity index (AI)	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-0.0632	-0.1000–0.0224	0.0010	0.0017
TerraClimate 1991–2020 climatic water deficit	Remove the pooled top 1% by area (26 units)	1,806/732	2000	900.2	758.3–1019.7	0.0010	0.0017
JRC 1984–2021 any observed surface-water fraction	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-0.0128	-0.0505–0.0375	0.5627	0.5627
Annual days with maximum UTCI ≥32 °C (oasis median)	Remove the pooled top 1% by area (26 units)	1,806/732	2000	140.2	109.9–166.7	0.0010	0.0010
Annual days with minimum UTCI ≤–13 °C (oasis median)	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-125.5	-154.0–69.63	0.0010	0.0010
Annual days with maximum UTCI ≥32 °C (2020 population weighted)	Remove the pooled top 1% by area (26 units)	1,806/732	2000	142.3	90.83–180.9	0.0010	0.0010

Metric	Analysis	n (BWh/BWk)	Valid replicates	Effect	95% interval	p	q
Annual days with minimum UTCI ≤ -13 °C (2020 population weighted)	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-96.71	-130.0—67.65	0.0010	0.0010
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2041–2070 relative to 1995–2014	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-0.180	-0.267—0.107	0.0010	0.0015
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2041–2070 relative to 1995–2014	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-0.160	-0.267—0.0426	0.0070	0.0093
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2041–2070 relative to 1995–2014	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-0.0298	-0.0417—0.0148	0.0010	0.0015
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2071–2099 relative to 1995–2014	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-0.237	-0.339—0.108	0.0010	0.0015
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2071–2099 relative to 1995–2014	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-0.226	-0.367—0.0326	0.0110	0.0132
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2071–2099 relative to 1995–2014	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-0.0460	-0.0611—0.0284	0.0010	0.0015
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2041–2070 relative to 1995–2014	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-0.230	-0.362—0.118	0.0010	0.0015
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2041–2070 relative to 1995–2014	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-0.216	-0.350—0.0197	0.0310	0.0338
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2041–2070 relative to 1995–2014	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-0.0444	-0.0579—0.0227	0.0010	0.0015
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2071–2099 relative to 1995–2014	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-0.521	-0.710—0.327	0.0010	0.0015
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2071–2099 relative to 1995–2014	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-0.228	-0.512—0.0331	0.1579	0.1579
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2071–2099 relative to 1995–2014	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-0.0816	-0.102—0.0481	0.0010	0.0015
FuturePop SSP2 oasis population, 2050	Remove the pooled top 1% by area (26 units)	1,806/732	2000	889.7	425.8–1421.2	0.0020	0.0020
FuturePop SSP2 oasis population, 2080	Remove the pooled top 1% by area (26 units)	1,806/732	2000	1046.7	554.4–1611.2	0.0010	0.0020
FuturePop SSP5 oasis population, 2050	Remove the pooled top 1% by area (26 units)	1,806/732	2000	801.9	374.9–1319.5	0.0010	0.0020
FuturePop SSP5 oasis population, 2080	Remove the pooled top 1% by area (26 units)	1,806/732	2000	847.9	391.8–1333.4	0.0020	0.0020
Kummu GDP v4 2020 total GDP	Remove the pooled top 1% by area (26 units)	1,806/732	2000	0.728	0.0542–1.31	0.0360	0.0480
Kummu GDP v4 2020 GDP per capita	Remove the pooled top 1% by area (26 units)	1,806/732	2000	-0.354	-0.664—0.0664	0.1239	0.1239
VIIRS Annual VNL v2.1 mean radiance	Remove the pooled top 1% by area (26 units)	1,806/732	2000	1.59	0.716–2.35	0.0020	0.0040
VIIRS Annual VNL v2.1 lit-area fraction	Remove the pooled top 1% by area (26 units)	1,806/732	2000	1.01	0.407–1.65	0.0010	0.0040
Global AI/ET0 v3.1 ET0	Remove the pooled top 1% by area (26 units)	1,806/732	2000	924.1	772.3–1065.7	0.0010	0.0017
Global AI/ET0 v3.1 ET0-SD	Remove the pooled top 1% by area (26 units)	1,806/732	2000	4.04	-7.32–14.72	0.5357	0.5627
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2041–2070 relative to 1995–2014	NEX polygon-only results	1,542/578	2000	-0.169	-0.248—0.0977	0.0010	0.0017
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2041–2070 relative to 1995–2014	NEX polygon-only results	1,542/578	2000	-0.149	-0.251—0.0423	0.0060	0.0080
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2041–2070 relative to 1995–2014	NEX polygon-only results	1,542/578	2000	-0.0287	-0.0419—0.0145	0.0010	0.0017
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2071–2099 relative to 1995–2014	NEX polygon-only results	1,542/578	2000	-0.219	-0.315—0.0998	0.0010	0.0017
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2071–2099 relative to 1995–2014	NEX polygon-only results	1,542/578	2000	-0.221	-0.348—0.0270	0.0210	0.0229
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2071–2099 relative to 1995–2014	NEX polygon-only results	1,542/578	2000	-0.0474	-0.0612—0.0278	0.0010	0.0017
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2041–2070 relative to 1995–2014	NEX polygon-only results	1,542/578	2000	-0.215	-0.335—0.101	0.0020	0.0030
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2041–2070 relative to 1995–2014	NEX polygon-only results	1,542/578	2000	-0.217	-0.338—0.0268	0.0180	0.0216
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2041–2070 relative to 1995–2014	NEX polygon-only results	1,542/578	2000	-0.0412	-0.0567—0.0225	0.0010	0.0017
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2071–2099 relative to 1995–2014	NEX polygon-only results	1,542/578	2000	-0.478	-0.671—0.316	0.0010	0.0017
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2071–2099 relative to 1995–2014	NEX polygon-only results	1,542/578	2000	-0.187	-0.494–0.0321	0.1499	0.1499
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2071–2099 relative to 1995–2014	NEX polygon-only results	1,542/578	2000	-0.0824	-0.104—0.0520	0.0010	0.0017
GHS-POP 2020 oasis population density	Area-weighted mean difference	1,822/742	NE	238.2	NE	NE	NE
GHS-BUILT-S 2020 built-up-area fraction	Area-weighted mean difference	1,822/742	NE	0.0016	NE	NE	NE
Background aridity index (AI)	Patch median difference	1,822/742	NE	-0.0474	NE	NE	NE
Background aridity index (AI)	Remove the top 1% by area	1,806/732	NE	-0.0632	NE	NE	NE
TerraClimate 1991–2020 climatic water deficit	Patch median difference	1,822/742	NE	809.3	NE	NE	NE
TerraClimate 1991–2020 climatic water deficit	Remove the top 1% by area	1,806/732	NE	900.2	NE	NE	NE
JRC 1984–2021 any observed surface-water fraction	Patch median difference	1,822/742	NE	-0.0008	NE	NE	NE
JRC 1984–2021 any observed surface-water fraction	Remove the top 1% by area	1,806/732	NE	-0.0128	NE	NE	NE
Annual days with maximum UTCI ≥ 32 °C (oasis median)	Population-weighted mean difference	1,822/742	NE	138.0	NE	NE	NE
Annual days with minimum UTCI ≤ -13 °C (oasis median)	Population-weighted mean difference	1,822/742	NE	-99.67	NE	NE	NE

Metric	Analysis	n (BWh/BWk)	Valid replicates	Effect	95% interval	p	q
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2041–2070 relative to 1995–2014	NEX polygon-only results	1,542/578	NE	-0.169	NE	NE	NE
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2041–2070 relative to 1995–2014	NEX polygon-only results	1,542/578	NE	-0.149	NE	NE	NE
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2041–2070 relative to 1995–2014	NEX polygon-only results	1,542/578	NE	-0.0287	NE	NE	NE
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmax, 2071–2099 relative to 1995–2014	NEX polygon-only results	1,542/578	NE	-0.219	NE	NE	NE
NEX-GDDP-CMIP6 SSP2-4.5 change in tasmin, 2071–2099 relative to 1995–2014	NEX polygon-only results	1,542/578	NE	-0.221	NE	NE	NE
NEX-GDDP-CMIP6 SSP2-4.5 change in pr, 2071–2099 relative to 1995–2014	NEX polygon-only results	1,542/578	NE	-0.0474	NE	NE	NE
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2041–2070 relative to 1995–2014	NEX polygon-only results	1,542/578	NE	-0.215	NE	NE	NE
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2041–2070 relative to 1995–2014	NEX polygon-only results	1,542/578	NE	-0.217	NE	NE	NE
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2041–2070 relative to 1995–2014	NEX polygon-only results	1,542/578	NE	-0.0412	NE	NE	NE
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmax, 2071–2099 relative to 1995–2014	NEX polygon-only results	1,542/578	NE	-0.478	NE	NE	NE
NEX-GDDP-CMIP6 SSP5-8.5 change in tasmin, 2071–2099 relative to 1995–2014	NEX polygon-only results	1,542/578	NE	-0.187	NE	NE	NE
NEX-GDDP-CMIP6 SSP5-8.5 change in pr, 2071–2099 relative to 1995–2014	NEX polygon-only results	1,542/578	NE	-0.0824	NE	NE	NE
FuturePop SSP2 oasis population, 2050	Group-median log ratio (no pseudocount)	1,822/742	NE	2.17	NE	NE	NE
FuturePop SSP2 oasis population, 2080	Group-median log ratio (no pseudocount)	1,822/742	NE	2.43	NE	NE	NE
FuturePop SSP5 oasis population, 2050	Group-median log ratio (no pseudocount)	1,822/742	NE	2.04	NE	NE	NE
FuturePop SSP5 oasis population, 2080	Group-median log ratio (no pseudocount)	1,822/742	NE	2.23	NE	NE	NE
Kummu GDP v4 2020 total GDP	Raw-scale patch median difference	1,822/742	NE	477589.5	NE	NE	NE
Kummu GDP v4 2020 GDP per capita	Raw-scale patch median difference	1,822/742	NE	-4092.0	NE	NE	NE
VIIRS Annual VNL v2.1 mean radiance	Raw-scale patch median difference	1,822/742	NE	0.442	NE	NE	NE
VIIRS Annual VNL v2.1 lit-area fraction	Raw-scale patch median difference	1,822/742	NE	0.272	NE	NE	NE
Global AI/ET0 v3.1 ET0	Patch median difference	1,822/742	NE	774.6	NE	NE	NE
Global AI/ET0 v3.1 ET0	Remove the top 1% by area	1,806/732	NE	924.1	NE	NE	NE
Global AI/ET0 v3.1 ET0-SD	Patch median difference	1,822/742	NE	2.64	NE	NE	NE
Global AI/ET0 v3.1 ET0-SD	Remove the top 1% by area	1,806/732	NE	4.04	NE	NE	NE

Panel C | FuturePop coverage and nearest-grid sensitivity

Scenario	Year	Variant	Sample treatment	n (BWh/BWk)	Effect (people)	95% interval	Two-sided plus-one p
SSP2	2050	V0	Primary analysis; all oases	1,822/742	887.62	440.56 to 1452.98	0.0030
SSP2	2050	V1	Exclude nearest-grid substitutions	1,692/657	1118.91	639.50 to 1655.75	0.0010
SSP2	2050	V2	Within-boundary coverage ≥ 0.50	1,278/397	1717.44	941.19 to 2526.88	0.0020
SSP2	2050	V3	Within-boundary coverage ≥ 0.95	582/118	1048.69	190.25 to 2085.84	0.0280
SSP2	2050	V4	Within-boundary coverage ≥ 0.99	362/71	531.81	-520.20 to 1456.69	0.2089
SSP2	2080	V0	Primary analysis; all oases	1,822/742	1052.04	575.19 to 1664.89	0.0010
SSP2	2080	V1	Exclude nearest-grid substitutions	1,693/657	1409.10	847.56 to 2006.55	0.0010
SSP2	2080	V2	Within-boundary coverage ≥ 0.50	1,281/397	2073.79	1243.14 to 3305.18	0.0010
SSP2	2080	V3	Within-boundary coverage ≥ 0.95	586/118	1451.22	397.71 to 2683.17	0.0200
SSP2	2080	V4	Within-boundary coverage ≥ 0.99	371/71	837.83	-514.85 to 1763.27	0.1369
SSP5	2050	V0	Primary analysis; all oases	1,822/742	829.91	385.37 to 1324.81	0.0020
SSP5	2050	V1	Exclude nearest-grid substitutions	1,692/657	1019.33	534.62 to 1553.18	0.0010
SSP5	2050	V2	Within-boundary coverage ≥ 0.50	1,279/397	1541.52	832.63 to 2368.00	0.0010
SSP5	2050	V3	Within-boundary coverage ≥ 0.95	582/118	987.65	125.56 to 1923.03	0.0340
SSP5	2050	V4	Within-boundary coverage ≥ 0.99	362/71	466.21	-671.65 to 1308.01	0.2649
SSP5	2080	V0	Primary analysis; all oases	1,822/742	868.66	409.12 to 1332.60	0.0010
SSP5	2080	V1	Exclude nearest-grid substitutions	1,693/657	1073.25	594.30 to 1592.78	0.0010
SSP5	2080	V2	Within-boundary coverage ≥ 0.50	1,281/397	1562.15	882.60 to 2712.10	0.0010
SSP5	2080	V3	Within-boundary coverage ≥ 0.95	585/118	1204.91	181.54 to 2129.58	0.0410
SSP5	2080	V4	Within-boundary coverage ≥ 0.99	368/71	696.28	-903.32 to 1446.76	0.2249

Note: Panel A examines 250-, 500- and 1,000-km spatial blocks; Panel B evaluates alternative estimands and sample treatments; and Panel C evaluates FuturePop coverage. In Panel C, V0 is the primary analysis, V1 excludes nearest-grid substitutions, and V2–V4 require within-boundary coverage of ≥ 0.50 , ≥ 0.95 and ≥ 0.99 , respectively. NE indicates that, for the corresponding sensitivity specification, the available output contained an effect estimate but no bootstrap replicate distribution; 95% intervals and p/q values therefore could not be reported. Positive effects indicate higher values in oases in hot-desert climates.

Supplementary Table 3 | Intermodel interquartile ranges for NEX-GDDP-CMIP6 future background-climate change across 30 models

Scenario	Period	Variable	Oasis class	n	Models	Median IQR	Unit
SSP2-4.5	2041–2070	Maximum-temperature change	Oases in hot-desert climates	1,822	30	0.574	°C
SSP2-4.5	2041–2070	Maximum-temperature change	Oases in cold-desert climates	742	30	0.717	°C
SSP2-4.5	2041–2070	Minimum-temperature change	Oases in hot-desert climates	1,822	30	0.577	°C
SSP2-4.5	2041–2070	Minimum-temperature change	Oases in cold-desert climates	742	30	0.627	°C
SSP2-4.5	2041–2070	Precipitation change	Oases in hot-desert climates	1,822	30	0.0713	mm d ⁻¹
SSP2-4.5	2041–2070	Precipitation change	Oases in cold-desert climates	742	30	0.0668	mm d ⁻¹
SSP2-4.5	2071–2099	Maximum-temperature change	Oases in hot-desert climates	1,822	30	0.759	°C
SSP2-4.5	2071–2099	Maximum-temperature change	Oases in cold-desert climates	742	30	0.849	°C
SSP2-4.5	2071–2099	Minimum-temperature change	Oases in hot-desert climates	1,822	30	0.926	°C
SSP2-4.5	2071–2099	Minimum-temperature change	Oases in cold-desert climates	742	30	1.06	°C
SSP2-4.5	2071–2099	Precipitation change	Oases in hot-desert climates	1,822	30	0.0707	mm d ⁻¹
SSP2-4.5	2071–2099	Precipitation change	Oases in cold-desert climates	742	30	0.0785	mm d ⁻¹
SSP5-8.5	2041–2070	Maximum-temperature change	Oases in hot-desert climates	1,822	30	0.755	°C
SSP5-8.5	2041–2070	Maximum-temperature change	Oases in cold-desert climates	742	30	0.907	°C
SSP5-8.5	2041–2070	Minimum-temperature change	Oases in hot-desert climates	1,822	30	0.744	°C
SSP5-8.5	2041–2070	Minimum-temperature change	Oases in cold-desert climates	742	30	0.958	°C
SSP5-8.5	2041–2070	Precipitation change	Oases in hot-desert climates	1,822	30	0.0853	mm d ⁻¹
SSP5-8.5	2041–2070	Precipitation change	Oases in cold-desert climates	742	30	0.0799	mm d ⁻¹
SSP5-8.5	2071–2099	Maximum-temperature change	Oases in hot-desert climates	1,822	30	1.26	°C
SSP5-8.5	2071–2099	Maximum-temperature change	Oases in cold-desert climates	742	30	1.55	°C
SSP5-8.5	2071–2099	Minimum-temperature change	Oases in hot-desert climates	1,822	30	1.57	°C
SSP5-8.5	2071–2099	Minimum-temperature change	Oases in cold-desert climates	742	30	1.90	°C
SSP5-8.5	2071–2099	Precipitation change	Oases in hot-desert climates	1,822	30	0.128	mm d ⁻¹
SSP5-8.5	2071–2099	Precipitation change	Oases in cold-desert climates	742	30	0.122	mm d ⁻¹

Note: Median IQR is the median across oases of the within-oasis interquartile range calculated from the 30-model ensemble and summarises intermodel dispersion. n denotes the number of oases.